LESSON PLAN

BASIC ELECTRICAL (23)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	02
Total No. of Lesson Plans	29	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Types of tanks, constructional details and arrangements in air frame.
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Types of tanks, constructional details and arrangements in air frame.
5. Specific Objectives	23.1.1.1 Trainee will be able to understand the construction and requirements of typical aircraft fuel system and its components.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Review of fuel tank diagrams from IB; Instructor supervision required for diagram interpretation.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary function of an aircraft fuel tank?
(b) Name one material commonly used for fuel tank construction.
(c) Why is fuel tank location critical in aircraft design?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Internal vs External Tank Types • Bladder Tank Construction Details • Integral Tank Arrangement in Wing • Fuel Tank Venting Systems • Sump and Drain Valve Locations	L & Ds	PPT, IB	15 min
(b)	• Fuel Tank Heating Elements • Anti-Skid and Anti-Slosh Devices • Tank Pressure Relief Valves • Fuel Tank Electrical Grounding • Corrosion Protection Measures	L & Ds	PPT, PDF	20 min
(c)	• Fuel Tank Filling Procedures • Emergency Fuel Dumping Systems • Tank Inspection and Maintenance • Fuel Quantity Indicator Calibration • Safety Precautions During Tank Work	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain the difference between an integral tank and a bladder tank.
(b) Describe the purpose of a fuel tank venting system.
(c) List three safety precautions when working near fuel tanks.

12. Home Assignment and Follow Up	Prepare a brief note on the classification of turbine engine fuels to be discussed in LP 03.
Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Fuel, major classification of fuel based on fuel state, turbine engine fuels	Refer to the master syllabus reference list	LP 03

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Instructor, Flight Lieutenant, P No. 12345	Senior Instructor, Wing Commander, P No. 67890
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min